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FITTED OPERATOR FINITE DIFFERENCE METHOD FOR SINGULARLY PERTURBED PARABOLIC CONVECTION-DIFFUSION TYPE

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ABSTRACT

In this paper, we study the numerical solution of singularly perturbed parabolic convection-diffusion type with boundary layers at the right side. To solve this problem, the backward-Euler with Richardson extrapolation method is applied on the time direction and the fitted operator finite difference method on the spatial direction is used, on the uniform grids. The stability and consistency of the method were established very well to guarantee the convergence of the method. Numerical experimentation is carried out on model examples, and the results are presented both in tables and graphs. Further, the present method gives a more accurate solution than some existing methods reported in the literature.

KEYWORDS

Singularly perturbation, convection-diffusion, boundary layer, Richardson extrapolation, fitted operator.

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PRACTICE OF CALIBRATION TECHNOLOGY FOR THE SPECIAL TEST EQUIPMENT

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ABSTRACT

For the issues encountered in the special test equipment calibration work, based on the characteristics of special test equipment, the calibration point selection, classification of calibration parameters and calibration method of special test equipment are briefly introduced in this paper, at the same time, the preparation and management requirements of calibration specification are described.

KEYWORDS:

Special Test Equipment, Calibration parameter, Calibration Method, Calibration specification

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LOW POWER SI CLASS E POWER AMPLIFIER AND RF SWITCH FOR HEALTH CARE

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ABSTRACT

This research was to design a 2.4 GHz class E Power Amplifier (PA) for health care, with 0.18um Semiconductor Manufacturing International Corporation CMOS technology by using Cadence software. And also RF switch was designed at cadence software with power Jazz 180nm SOI process. The ultimate goal for such application is to reach high performance and low cost, and between high performance and low power consumption design. This paper introduces the design of a 2.4GHz class E power amplifier and RF switch design. PA consists of cascade stage with negative capacitance. This power amplifier can transmit 16dBm output power to a 50Ω load. The performance of the power amplifier and switch meet the specification requirements of the desired.

KEYWORDS

Cascode, Negative Capacitance, Class E, Power amplifier, Healthcare, RF switch

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RESEARCH IN BIG DATA – AN OVERVIEW

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ABSTRACT

Big data is a prominent term which characterizes the improvement and availability of data in all three formats like structure, unstructured and semi formats. Structure data is located in a fixed field of a record or file and it is present in the relational data bases and spreadsheets whereas an unstructured data file includes text and multimedia contents. The primary objective of this big data concept is to describe the extreme volume of data sets i.e. both structured and unstructured. It is further defined with three "V" dimensions namely Volume, Velocity and Variety, and two more "V" also added i.e. Value and Veracity. Volume denotes the size of data, Velocity depends upon the speed of the data processing, Variety is described with the types of the data, Value which derives the business value and Veracity describes about the quality of the data and data understandability. Nowadays, big data has become unique and preferred research areas in the field of computer science. Many open research problems are available in big data and good solutions also been proposed by the researchers even though there is a need for development of many new techniques and algorithms for big data analysis in order to get optimal solutions. In this paper, a detailed study about big data, its basic concepts, history, applications, technique, research issues and tools are discussed.

KEYWORDS

Big data, Technologies, Visualization, Classification, Clustering

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THE PRESSURE SIGNAL CALIBRATION TECHNOLOGY OF THE COMPREHENSIVE TEST SYSTEM

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ABSTRACT

The pressure signal calibration technology of the comprehensive test system which involved pressure sensors was studied in this paper. The melioration of pressure signal calibration methods was elaborated. Compared with the calibration methods in the lab and after analyzing the relevant problems, the calibration technology online was achieved. The test datum and reasons of measuring error analyzed, the uncertainty evaluation was given and then this calibration method was proved to be feasible and accurate.

KEYWORDS

Calibration online, Pressure transducer, Pressure signal calibration.

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CLASS D POWER AMPLIFIER FOR MEDICAL APPLICATION

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ABSTRACT

The objective of this research was to design a 2.4 GHz class AB Power Amplifier (PA), with 0.18um Semiconductor Manufacturing International Corporation (SMIC) CMOS technology by using Cadence software, for health care applications. The ultimate goal for such application is to minimize the trade-offs between performance and cost, and between performance and low power consumption design. This paper introduces the design of a 2.4GHz class D power amplifier which consists of two stage amplifiers. This power amplifier can transmit 15dBm output power to a 50Ω load. The power added efficiency was 50% and the total power consumption was 90.4 mW. The performance of the power amplifier meets the specification requirements of the desired.

KEYWORDS

Two stage, Class D, Power amplifier, Healthcare.

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AN IMPROVED METHOD TO DETECT INTRUSION USING MACHINE LEARNING ALGORITHMS

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ABSTRACT

An intrusion detection system detects various malicious behaviors and abnormal activities that might harm security and trust of computer system. IDS operate either on host or network level via utilizing anomaly detection or misuse detection. Main problem is to correctly detect intruder attack against computer network. The key point of successful detection of intrusion is choice of proper features. To resolve the problems of IDS scheme this research work propose "an improved method to detect intrusion using machine learning algorithms". In our paper we use KDDCUP 99 dataset to analyze efficiency of intrusion detection with different machine learning algorithms like Bayes, NaiveBayes, J48, J48Graft and Random forest. To identify network based IDS with KDDCUP 99 dataset, experimental results shows that the three algorithms J48, J48Graft and Random forest gives much better results than other machine learning algorithms. We use WEKA to check the accuracy of classified dataset via our proposed method. We have considered all the parameter for computation of result i.e. precision, recall, F – measure and ROC.

KEYWORDS

IDS, KDDCUP 99, Machine learning, WEKA, Network Security, Precision, Recall.

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ADAPTIVE MODELING OF URBAN DYNAMICS DURING EPHEMERAL EVENT VIA MOBILE PHONE TRACES

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ABSTRACT

The communication devices have produced digital traces for their users either voluntarily or not. This type of collective data can give powerful indications that are affecting the urban systems design and development. In this study mobile phone data during Armada event is investigated. Analyzing mobile phone traces gives conceptual views about individuals densities and their mobility patterns in the urban city. The geo-visualization and statistical techniques have been used for understanding human mobility collectively and individually. The undertaken substantial parameters are inter-event times, travel distances (displacements) and radius of gyration. They have been analyzed and simulated using computing platform by integrating various applications for huge database management, visualization, analysis, and simulation. Accordingly, the general population pattern law has been extracted. The study contribution outcomes have revealed both the individuals densities in static perspective and individuals mobility in dynamic perspective with multi levels of abstraction (macroscopic, mesoscopic, microscopic).

KEYWORDS

Modeling, urban, mobility, Armada, inter-event time, radius of gyration, travel distance, trajectory, CDRs.

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CLASSIFIER SELECTION MODELS FOR INTRUSION DETECTION SYSTEM (IDS)

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ABSTRACT

Any abnormal activity can be assumed to be anomalies intrusion. In the literature several techniques and algorithms have been discussed for anomaly detection. In the most of cases true positive and false positive parameters have been used to compare their performance. However, depending upon the application a wrong true positive or wrong false positive may have severe detrimental effects. This necessitates inclusion of cost sensitive parameters in the performance. Moreover the most common testing dataset KDD-CUP-99 has huge size of data which intern require certain amount of pre-processing. Our work in this paper starts with enumerating the necessity of cost sensitive analysis with some real life examples. After discussing KDD-CUP-99 an approach is proposed for feature elimination and then features selection to reduce the number of more relevant features directly and size of KDD-CUP-99 indirectly. From the reported literature general methods for anomaly detection are selected which perform best for different types of attacks. These different classifiers are clubbed to form an ensemble. A cost opportunistic technique is suggested to allocate the relative weights to classifiers ensemble for generating the final result. The cost sensitivity of true positive and false positive results is done and a method is proposed to select the elements of cost sensitivity metrics for further improving the results to achieve the overall better performance. The impact on performance trade of due to incorporating the cost sensitivity is discussed.

KEYWORDS

Intrusion detection system (IDS), True positive (TP), False Positive (FP), Support Vector Machine (SVM).

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FROM FPGA TO ASIC IMPLEMENTATION OF AN OPENRISC BASED SOC FOR VOIP APPLICATION

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ABSTRACT

ASIC (Application Specific Integrated Circuit) design verification takes as long as the designers take to describe, synthesis and implement the design. The hybrid approach, where the design is first prototyped on an FPGA (Field-Programmable Gate Array) platform for functional validation and then implemented as an ASIC allows earlier defect detection in the design process and thus allows a significant time saving. This paper deals with a CMOS standard-cell ASIC implementation of a SoC (System on Chip) based on the Open RISC processor for Voice over IP (VoIP) application; where a hybrid approach is adopted. The architecture of the design is mainly based on the reuse of IPs cores described at the RTL level. This RTL code is technology-independent; hence the design can be ported easily from FPGA to ASIC. Results show that the SoC occupied the area of 2.64mm². Regarding the power consumption, RTL power estimation is given.

KEYWORDS

ASIC, FPGA, Opencores, OpenRISC, SoC, VoIP

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